

Assignment #8

Finite State Machines

Practical Application of FSM (Mealy)

1. **Detect sequence 1011 in a serial bit stream.** *Hint: Use a 4-state FSM and output 1 when the full sequence is matched. (overlap allowed)*
2. **Detect sequence 110 with overlap allowed.** *Hint: Track overlapping by transitioning back to state for last 1.*
3. **Detect alternating pattern 101010...** *Hint: Toggle between expecting 1 and 0 alternately.*
4. **Detect any sequence ending with 1001.** *Hint: Maintain a 4-bit shift history in state transitions. (Overlapping Not Allowed)*
5. **Detect three consecutive 1's in a stream.** *Hint: Use states to count how many 1's have appeared in a row.*
6. **Detect pattern 0110 used in bit-stuffing protocols.** *Hint: Design FSM to output 1 upon detecting the full pattern.*
7. **Detect two 0's followed by a 1 (i.e., 001).** *Hint: Use 3 states to monitor input history.*
8. **Detect binary palindrome pattern (e.g., 1001, 0110).** *Hint: Track mirrored positions via FSM logic.*

State Transition Tables for Mealy FSM Exercises

1. Detect sequence 1011

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S2, 0)	(S1, 0)
S2	(S0, 0)	(S3, 0)
S3	(S2, 0)	(S1, 1)

2. Detect sequence 110 (with overlap)

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S0, 0)	(S2, 0)
S2	(S0, 1)	(S2, 0)

3. Detect alternating pattern 101010...

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S0, 1)	(S1, 0)

4. Detect sequence ending with 1001

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S2, 0)	(S1, 0)
S2	(S3, 0)	(S1, 0)
S3	(S0, 0)	(S0, 1)

5. Detect three consecutive 1's

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S0, 0)	(S2, 0)
S2	(S0, 0)	(S0, 1)

6. Detect pattern 0110

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S1, 0)	(S0, 0)
S1	(S1, 0)	(S2, 0)
S2	(S1, 0)	(S3, 0)
S3	(S0, 1)	(S2, 0)

7. Detect sequence 001

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S1, 0)	(S0, 0)
S1	(S2, 0)	(S0, 0)
S2	(S2, 0)	(S0, 1)

8. Detect binary palindrome (e.g., 1001)

Present State	Next State,Output	
	Input = 0	Input = 1
S0	(S0, 0)	(S1, 0)
S1	(S2, 0)	(S1, 0)
S2	(S3, 0)	(S2, 0)
S3	(S0, 1)	(S3, 0)

Practical Application of FSM (Moore)

1. **Detect sequence 1011 in a serial bit stream.** *Hint: Use a 5-state FSM. Output = 1 is associated only with the final state. (overlap allowed)*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S2	S1
S2	0	S0	S3
S3	0	S2	S4
S4	1	S2	S1

2. **Detect sequence 110 with overlap allowed.** *Hint: Use 4 states and set output=1 only in final state.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S0	S2
S2	0	S3	S2
S3	1	S0	S2

3. **Detect alternating pattern 101010...** *Hint: Output = 1 when expected bit is matched.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	1	S2	S0
S2	1	S0	S1

4. **Detect any sequence ending with 1001.** *Hint: Use 5 states, output = 1 on final state.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S2	S1
S2	0	S3	S1
S3	0	S0	S4
S4	1	S2	S1

5. **Detect three consecutive 1's in a stream.** *Hint: Output 1 when three 1s seen continuously.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S0	S2
S2	0	S0	S3
S3	1	S0	S3

6. Detect pattern 0110 used in bit-stuffing protocols. *Hint: Output = 1 at final state.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S0	S2
S2	0	S3	S0
S3	1	S0	S1

7. Detect two 0's followed by a 1 (i.e., 001). *Hint: 3 states are sufficient.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S1	S0
S1	0	S2	S0
S2	0	S2	S3
S3	1	S1	S0

8. Detect binary palindrome pattern (e.g., 1001). *Hint: FSM tracks mirrored states.*

Present State	Output	Next State	
		Input = 0	Input = 1
S0	0	S0	S1
S1	0	S2	S1
S2	0	S3	S2
S3	0	S0	S4
S4	1	S1	S1

Assignment

1. Design a Finite-State Machine (FSM) that generates a 3-bit Gray Code sequence on each clock cycle.
2. Design an FSM based 3-bit even parity bit checker. Extend it to 4-bit after a successful implementation of 3 bit.
3. **Parking Payment System** : Allow entry when total coins inserted ≥ 20 .
Inputs: 5, 10, 20 coins
States: Represent total amount inserted
Output:
 - **0:** Gate closed
 - **1:** Gate opens (only in 20 state)
4. Design an FSM for a vending machine that sells an item for 10. The user can insert coins of Rs. 1, 2, or 5.
States: 0, 1, 2, 5, 7, 10
Inputs: Coin values: 1, 2, 5
Output: Dispense item when 10 is reached.

Self Learning Assignment

1. Design Subtractor using Mealy and Moore sequential circuit
2. Design a 3-bit Gray code counter
3. Design Parity Bit Generator using Mealy Sequential Circuits
4. Design Mealy sequential circuit for detecting 000 or 111 (Overlapping is allowed)
5. Design Mealy sequential circuit for the detecting input sequence 1100, 1010 or 1001. (The circuit resets to its initial state after a 1 output has been generated. - Overlapping Not Allowed)
6. Design Mealy sequential circuit for the detecting input sequence 1101, 1011 or 1001. (The circuit resets to its initial state after a 1 output has been generated. - Overlapping Not Allowed)